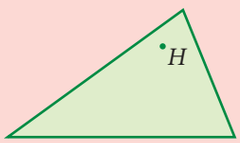
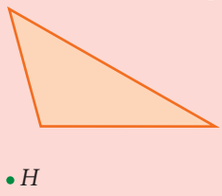
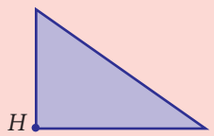
Note

Where do the Orthocentre lie in the given triangles.

	Acute Triangle	Obtuse Triangle	Right Triangle
Orthocentre	Inside of Triangle 	Outside of Triangle 	Vertex at Right Angle 

Exercise 4.5

1. Construct the ΔLMN such that LM=7.5cm, MN=5cm and LN=8cm. Locate its centroid.
2. Draw and locate the centroid of the triangle ABC where right angle at A, AB = 4cm and AC = 3cm.
3. Draw the ΔABC, where AB = 6cm, $\angle B = 110°$ and AC = 9cm and construct the centroid.
4. Construct the ΔPQR such that PQ = 5cm, PR= 6cm and $\angle QPR = 60°$ and locate its centroid.
5. Draw ΔPQR with sides PQ = 7 cm, QR = 8 cm and PR = 5 cm and construct its Orthocentre.
6. Draw an equilateral triangle of sides 6.5 cm and locate its Orthocentre.
7. Draw ΔABC, where AB = 6 cm, $\angle B = 110^0$ and BC = 5 cm and construct its Orthocentre.
8. Draw and locate the Orthocentre of a right triangle PQR where PQ = 4.5 cm, QR = 6 cm and PR = 7.5 cm.

4.7.3 Construction of the Circumcentre of a Triangle

Circumcentre

The Circumcentre is the point of concurrency of the Perpendicular bisectors of the sides of a triangle.
It is usually denoted by S.

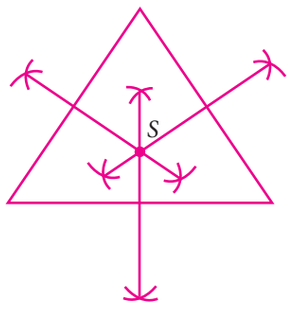


Fig. 4.90

Circumcircle

The circle passing through all the three vertices of the triangle with circumcentre (S) as centre is called circumcircle.

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